

RK200-03 Pyranometer



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The RK200-03 Pyranometer is produced based thermoelectric principle; sensing elements are made by winding - plated thermopiles with multi contacts. Its surface is coated by black coating with high absorption rate. Hot contacts on the sensors surface, while the cold junction is located within the body, temperature difference between the hot and cold junction generates electromotive force, the thermoelectric effect is proportional to the solar radiation. In order to reduce the ambient temperature effect temperature compensation circuit designed here to reduce the effects to products properties.

FEATURES

- Conform to the WMO standard
- Suitable for harsh environment
- High sensitivity
- Measuring without power
- Easy installation

Parts

1. Sensor with screws: 1
2. Protection cover:
3. Cable: 1
4. Aluminum alloy instrument box: 1

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SPECIFICATIONS

Item	SPECIFICATIONS
Spectral range	300~3200nm
Supply	Mark on the label
Range	0-2000W/m ²
Output	0-20mV,0-2.5V,0-5V,4-20mA,RS485
Sensitivity	7-14μV*W ⁻¹ *m ²
Internal resistance	350Ω
Response time	≤20s(99%)
Cosine correction	≤±7%(Solar elevation angle=10°)
Measuring angle	2π solid angle
Non-linear	≤±2%
Temperature effect	±2%(-10℃-+40℃)
Stability	≤±2%/year
Operating Temperature	-40℃-+60℃
Ingress Protection	IP65
Weight(unpacked)	2.5kg
Dimension	Ø165*120mm
Shell material	Aluminum alloy
Storage Condition	10℃-60℃@20%-90%RH

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① If you choose to 0-20 mv signal output, without power supply, if you choose to 0-2.5 v output, supply is 5VDC.

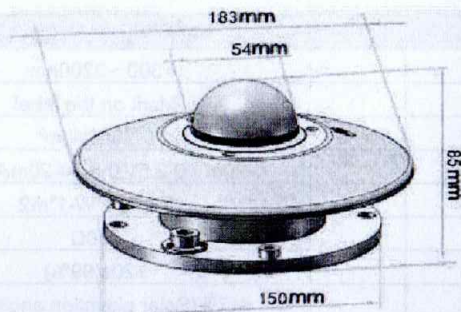
CALCULATION METHOD

- **0-20mV**
Solar radiation values(W/m²)= Voltage output value(μV)/sensitivity coefficient(μV*W⁻¹*m²), Each product is with one sensitivity coefficient respectively (It is mentioned on the product's label)
- **0-2.5V**
Solar radiation values(W/m²)=(V/2.5)*2000(Where V is output voltage value,unit:V)
- **0-5V**
Solar radiation values(W/m²)=(V/5)*2000(Where V is output voltage value,unit:V)
- **4-20mA**
Solar radiation values(W/m²)=(I-4)/16*2000(Where I is output current value,unit:mA)
- **RS485**

MODBUS-RTU

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DIMENSION



Installation hole distance:120mm

ELECTRICAL CONNECTIONS

Cable	Voltage	Current	RS485	mV
Red (Pin1)	V+	V+	V+	Vout+
Yellow (Pin4)	Vout	Iout	RS485A	Vout-
Black/Green (Pin2)	V-	V-	V-	
Blue (Pin5)			RS485B	

MOUNTING & MAINTENANCE

- The sensor should be installed in the open air without any cover above the sensing surface.
- The sensor connector should be faced to the north, fix it after the horizontal position is well adjusted.
- Please check the filter cover regularly & make sure it is clean.
- Please do not remove or loose the filter cover,otherwise the accuracy will be affected.
- Please make sure the desiccant to be dry .(If the color of desiccant is changed from orange to green, it should be replaced,it is recommended to replace the desiccant every 6 month.)
- Protection cover is not necessary in general rainfall, but if prolonged heavy rains or hail, the protective cover is recommended to be installed.
- The sensitivity is recommended to be re-calibrated after two year use.

Communication Protocol (MODBUS)

Transmission mode: MODBUS-RTU,Baud rate: 9600bps,Data bits:8,Stop bit:1,Check bit:no

Slave address:the factory default is 01H (set according to the need,00H to FFH)

- The 03H Function Code Example: Read The Radiation Value

Host Scan Order(slave address:0x01)

01 03 00 00 00 01 840A

Slave Response

01 03 02 03 B4 B8C3

Solar Radiation:(03B4)H=(948)D=948(W/m²)

- The 10H Function Code Example: Modify the slave address

Host Scan Order (Changed to 01H,read and write address must be 00H):

00 10 01 BDC0

Slave Response:

00 10 007C

- The 20H Function Code Example: Read The Slave Address(fixed command,ensure that no other devices on the bus)

Host Scan Order:

00 20 00 68

Slave Response (addr. =01H):

00 20 01 A9C0

Note:

1. All underlined is fixed bit;
2. The last two bytes is CRC check command.

WARRANTY

This product is warranted to be free of defects in materials and construction for a period of 12 months from date of lead time. Liability is limited to repair or replacement of defective item.

CE Complies with applicable CE directives.

Manual subject to change without notice. Version 1.0